

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
6 FT 1 HT	(a)			$3000 - 600 = 2400$ (1) $\frac{120000}{2400}$ (1) (ecf) $= 50$ months(1) $\frac{120000}{600} / \frac{120000}{3000} = (0)$ marks		3		3	3	
	(b)			Ay $2 \times (1)$ from: wind power is intermittent (1) noise (1) spoils view (1) more land needed (1) kills birds (1)	2			2		
	(c)	(i)		$\frac{160}{480}$ kWh $\times 100$ (1) (selection and substitution) $= 33(.33\%)$ (1) Accept 0.33 for (1)		2		2	2	
		(ii)		His cows produce: 120×3 (1) $= 360$ tonnes of waste (1) He needs: $\frac{160000}{160}$ (1) $= 1000$ tonnes so he doesn't have enough cows/less than his cows produce(1)			4	4	4	

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				OR He needs: $\frac{160000}{160} \text{ (1)}$ =1000 tonnes (1) No of cows = $\frac{1000}{3} \text{ (1)}$ =333.3 so he hasn't got enough cows (1) OR 120 × 3 = 360 tonnes of waste. 1 tonne 160kWh so 360 × 160 = 57 600kWh electricity he needs 160 000 kWh so not enough electrical energy produced to run his farm OR 160 × 3 (1) = 480 kWh per cow (1) 120 × 480 = 57 600 kWh (1) 160 000 – 57 600 = 102 400 extra needed so not enough (1) Accept conclusion based on candidate answer ecf						
				Question 6 total	2	5	4	11	9	0